Full Stack Developer Challenge: Walkie Talkie Solution

Objective

Your challenge is to design and build a walkie talkie solution, leveraging your strengths and demonstrating your creativity. The assignment is intentionally open-ended to allow you to take the initiative and show us how you shine in your area of expertise.

The Challenge

You are tasked with creating a walkie talkie application. It could be a beautifully interactive front-end UI, a robust back-end streaming service, or a full-stack end-to-end system - the choice is yours.

Your focus could be on:

- **Front-end**: Design a visually appealing, intuitive, and interactive UI for users to communicate.
- **Back-end**: Build a strong, scalable, and reliable streaming service that can handle multiple simultaneous connections.
- **Full-stack**: Combine the best of both worlds. Create an end-to-end system that provides a seamless user experience from interface to server.

Regardless of your focus, we expect a concept for the overall system. Provide a conceptual framework for the parts of the system you don't get to. Show us your thought process, your design and architecture skills, and how you would scale the system to handle a larger user base.

Expectations

The challenge is designed to mimic a real work environment:

1. **Creativity**: This assignment doesn't have a singular correct solution. We are excited to see your unique approach.

2. **Conceptual thinking**: If you can't build the entire system, we'd still like to understand your vision for it.
3. **Communication**: Be ready to explain your design decisions, your approach to scalability and robustness, and your thought process.
4. **Collaboration**: Just like in a real work scenario, you're free to use any resources at your disposal. "Cheating" is allowed - feel free to use libraries, frameworks, Stack Overflow, etc. We're more interested in how you solve problems than in testing your memory.
5. **Sprinkle some LLM stuff in here:** We don’t care what exactly it is, but show us you know how to use LLMs as a tool and explore a concept that could work in the age of AI.

Timeframe

You have 5 hours to complete this challenge. We're not just looking for what you complete in this time but also how you manage it.

Submission

Send us a link to your Github repo.

We look forward to seeing your unique solutions! Good luck!